

SIMIYU KEVIN MUKUBWA

kevsim98@uw.edu — +12063352768, +254712014216

LinkedIn Profile — Personal Website

SUMMARY

Lecturer in the Department of Mathematics at the University of Nairobi, teaching statistics and applied mathematics, with a PhD proposal in transcriptomics and Bayesian hierarchical modeling of HIV host factors under review. Master of Science Biostatistics student(University of Washington) interested in using evidence-based, data driven interventions to address critical health challenges in public health, clinical trials and statistical methods in spatial epidemiology. Proficient in analyzing and interpreting complex data sets, and skilled in data cleaning, analysis and inferential statistics and model selection.

SKILLS

- **Study Design & Research Process:** Hypothesis formation → study design → data collection protocols → Statistical Analysis Plan (SAP) → analysis → interpretation → communication of findings (cross-sectional and longitudinal studies); cross-sectional analysis; clinical trial design and analysis; sample size and power calculations (cross-sectional and longitudinal studies); epidemiologic study designs (case-control, cohort, cross-sectional)
- **Foundational Statistics:** Descriptive statistics (summary measures, exploratory data analysis); statistical inference and hypothesis testing (parametric and non-parametric)
- **Statistical Methods:**
 - *Longitudinal/multilevel data analysis:* linear mixed effects models and GEE
 - *Survival analysis:* Cox regression, RMST, competing risks (cause-specific hazards, subdistribution Fine-Gray, CIFs)
 - *Bayesian statistics:* priors, MCMC methods (Metropolis, Gibbs, HMC/NUTS), Bayesian joint modeling of longitudinal and time-to-event outcomes, Bayesian Kernel Machine Regression (BKMR) for mixed-exposure analysis
 - *Spatial epidemiology:* small area estimation using survey/DHS data
- **Computational Tools:** R, Python, SQL, Excel, Tableau, GitHub; INLA (spatial/survey modeling); Stan/CmdStan (Bayesian modeling); LaTeX/Beamer (PDF, Word, PPT and slide creation)
- **Genomics, Transcriptomics & Bioinformatics (developing):** scRNA-seq analysis (Seurat, pseudobulk DESeq2/edgeR, trajectory analysis via Slingshot, multi-modal WNN integration, GSEA); scATAC-seq analysis; spatial transcriptomics; NGS quality control pipelines (sequence quality, alignment, mapping quality)
- **Interpersonal Skills:** Cross-disciplinary collaboration (clinicians, epidemiologists, laboratory scientists); international collaboration (Kenya and the US); communicating statistical findings to non-statistical audiences; mentoring and teaching; teamwork; time management; problem-solving.
- **Languages:** English

EDUCATION

University of Nairobi, Faculty of Science & Technology	PhD proposal under review
<i>Doctor of Philosophy (PhD) – proposed</i>	

University of Washington, School of Public Health, Seattle WA US	
<i>MSc in Biostatistics (Capstone) program</i>	Current GPA: 3.75/4.0

University of Nairobi, Nairobi, Kenya	09/2020 – 06/2022
<i>MSc in Biometry</i>	GPA: 3.8/4.0

University of Nairobi, Nairobi, Kenya	01/2017 – 06/2020
<i>BEd (Science) Mathematics major</i>	
Graduated with First Class Honors 09/2020	

EXPERIENCE

Work Experience

Fred Hutchinson Cancer Center, Seattle, WA, USA

Volunteer Biostatistician

06/2026 – 09/2026

- Supported the statistical analysis of ATORTUB, a multi-site, randomized Phase IIC trial of adjunctive atorvastatin (20, 40 and 60 mg) added to standard anti-tuberculosis therapy in drug-susceptible pulmonary TB.
- Analyzed microbiological (sputum culture conversion), radiological and lung function outcomes in R, and prepared analysis tables and figures for the trial report.

University of Nairobi, Nairobi, Kenya

Lecturer, Faculty of Science and Technology (Mathematics Department)

12/2022 – to present

- Taught statistics courses such as Introduction to statistics, statistical inference, applied statistics, fundamental of engineering mathematics, Geospatial engineering, Civil and mechanical engineering, Introduction to Optimization and operation Research to over 3,000 number of students.
- Developed and administered exams and tutorials, graded assignments, and developed lectures.
- Collaborated with over 50 professors from ten different faculties, faculty of education(science), faculty of arts, faculty of economic & statistics, faculty of civil & mechanical engineering, Geospatial and Bio-system engineering.
- Helped students realize their academic goals by providing mentoring students via office hours, 1:1 counseling sessions and academic advice.

Research Projects

1. Determinants of Contraceptive Method Choices Among Married Muslim Women in Indonesia

- **Description:** Conducted a cross-sectional study analyzing data from the Demographic and Health Survey (DHS) to examine socio-demographic determinants influencing contraceptive choices.
- **Data Skills Demonstrated:** Data cleaning and preparation of large survey datasets; Multinomial logistic regression modeling; Statistical inference (odds ratios, confidence intervals, p-values); Interpretation of complex categorical data; Visualization and reporting of results for publication in Bulletin of Mathematics and Statistics Research (Vol. 12, Issue 2, 2024).

2. A Geospatial Comparative Analysis of SAE Models for Antenatal Care Coverage in Kenya

Conducted small area estimation using spatial models (Fay Herriot, unstratified and stratified cluster level) to find the smoothed estimates for the coverage of ANC(4+) in subnational regions in Kenya leveraging the borrowing strength.

3. Long-Term CD4 Recovery Patterns and Cofactors in Children with Perinatally Acquired HIV

Secondary analysis of the Optimizing Pediatric HIV (OPH) cohort in Nairobi, Kenya (104 children with perinatally acquired HIV, up to 15 years post-ART) using linear and piecewise mixed effects models to characterize CD4% trajectories and their predictors. CD4% recovery was biphasic (+2.11 points/year in the first 2 years, +0.62 thereafter); higher baseline CD4%, female sex, and early viral suppression predicted higher CD4% throughout follow-up.

4. Longitudinal Lipid Trajectories in Kenyan Children with Perinatally Acquired HIV

Secondary analysis of the Optimizing Pediatric HIV cohort (158 children, 494 repeated lipid measurements through age 10) using modified Poisson regression with robust standard errors and GEE to characterize lipid trajectories and early-life correlates of dyslipidemia.

5. Systematic, Age- and Strain-Stratified Identification of Pro- and Anti-HIV Host Factor Genes in CD4+ T Cells of HIV-Infected and Uninfected Children and Adults in Kenya: A Transcriptomic and Bayesian Hierarchical Modeling Framework

PhD research proposal (under review) to identify pro- and anti-HIV host factor genes in CD4+ T cells, stratified by age and viral strain, comparing HIV-infected and uninfected children and adults in Kenya using transcriptomics and Bayesian hierarchical modeling.

PUBLICATIONS

Published

- **Simiyu K**, Muhua G. Multinomial Logistic Regression to Study the Determinants of Choices of Contraceptive Methods among Married Muslim women in Indonesia. *Bulletin of Mathematics and Statistics Research (BOMSR)* Volume 12, Issue 2; 2024. <http://www.bomsr.com>

Manuscripts in Preparation

- **Simiyu K**, Neary J, Wamalwa D, Chebet D, Ojee E, Benki-Nugent S, Lehman D, Slyker JA, Owiti P, Richardson B, John-Stewart G. Long-Term CD4 Recovery Patterns and Cofactors in Children with Perinatally Acquired HIV. *Manuscript in preparation*.
- Ojee E, Neary J, Wamalwa D, Chebet D, **Simiyu K**, Benki-Nugent S, Lehman D, Slyker JA, Owiti P, Richardson B, John-Stewart G. Longitudinal Lipid Trajectories in Kenyan Children with Perinatally Acquired HIV. *Manuscript in preparation*.